	Scene	Atrium Night						Concourse Night					
	Camera	Pinhole	Fisheye1	Fisheye2	Fisheye3	360	Average	Pinhole	Fisheye1	Fisheye2	Fisheye3	360	Average
APR	PN [1]	19.8/90.8	16.4/87.3	16.1/86.7	15.7/83.5	8.7/22.8	15.3/74.2	19.8/96.2	13.3/85.5	13.0/88.4	11.1/87.5	5.5/45.6	12.5/80.6
APR	MS-T [1]	24.8/84.7	17.8/80.5	17.0/80.1	15.1/80.0	5.0/46.3	15.9/74.3	22.1/74.7	17.7/69.5	17.0/68.3	17.2/72.7	8.4/57.3	16.5/68.5
APR	PN-VC2 [1]	13.6/50.2	11.5/38.1	11.6/38.3	11.4/37.8	9.7/45.7	11.6/42.0	10.7/25.4	8.3/16.4	8.4/15.7	8.1/14.7	5.5/13.5	8.2/17.1
APR	MS-T-VC2 [1]	13.5/70.6	9.1/64.1	9.0/63.8	8.6/64.4	7.3/61.5	9.5/64.9	5.5/28.7	4.0/20.1	3.9/20.3	3.5/20.9	3.4/23.7	4.1/22.7
APR	CDG-Loc [2]	4.6/25.3	**1.8**/11.0	**1.7**/10.1	**1.6**/10.1	**1.4**/7.3	**2.2**/12.8	3.3/11.4	1.7/6.6	1.6/6.3	1.4/6.9	**1.1**/6.1	1.8/7.5
SCR	Ray-ACEG (Ours)	**4.2/10.9**	2.4/**6.5**	2.3/**6.2**	2.2/**6.5**	2.1/**6.0**	2.7/**7.2**	**1.7/8.0**	**1.2/5.8**	**1.2/5.9**	**1.2/5.5**	1.2/**5.0**	**1.3/6.0**
	Scene	Hall Night						Piatrium Night					
	Camera	Pinhole	Fisheye1	Fisheye2	Fisheye3	360	Average	Pinhole	Fisheye1	Fisheye2	Fisheye3	360	Average
APR	PN [1]	22.8/95.6	18.3/93.8	17.9/93.6	16.8/91.3	10.9/100.1	17.3/94.9	33.2/89.7	29.3/89.6	28.8/90.4	29.2/89.8	30.6/98.5	30.2/91.6
APR	MS-T [1]	32.1/96.1	29.9/94.2	29.7/93.6	31.8/93.1	22.3/94.4	29.2/94.3	37.7/91.3	37.3/89.1	36.2/89.0	35.7/85.6	37.2/83.0	36.8/87.6
APR	PN-VC2 [1]	13.1/56.1	10.5/43.7	10.1/43.0	9.7/40.5	7.5/34.7	10.2/43.6	15.8/51.9	13.2/34.9	12.8/34.8	12.5/32.6	10.6/30.9	13.0/37.0
APR	MS-T-VC2 [1]	8.9/73.1	5.6/67.6	5.3/66.3	4.8/68.9	3.8/67.2	5.7/68.6	16.0/63.9	13.0/53.8	12.4/53.7	11.9/49.9	11.9/49.9	13.0/54.2
APR	CDG-Loc [2]	4.2/32.5	2.2/18.5	2.1/17.5	1.9/15.7	1.5/18.0	2.4/20.4	7.2/27.9	4.6/16.1	4.3/16.1	3.7/14.1	**2.6**/9.6	4.5/16.8
SCR	Ray-ACEG (Ours)	**2.6/12.1**	**1.6/6.8**	**1.6/7.3**	**1.6/7.0**	**1.1/5.2**	**1.7/7.7**	**6.2/13.2**	**3.3/7.1**	**3.2/7.1**	**3.1/7.2**	2.7/**8.5**	**3.7/8.6**